

Foto: Dr. Matthias Otte

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EUROfusion

HELMHOLTZ

RESEARCH FOR GRAND CHALLENGES



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IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

February 26, 2026

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MOTIVATION AND CONTEXT

NUCLEAR FUSION: THE ENERGY CHALLENGE

- D-T fusion reaction: ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He}$
(3.5 MeV) + n (14.1 MeV)
- Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

- Requires: high temperature ($T \sim 10 \text{ keV}$ to 20 keV), density, and confinement time
- Magnetic confinement: tokamaks and stellarators

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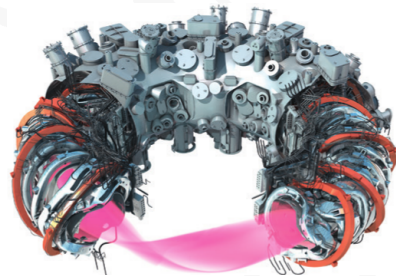
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Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

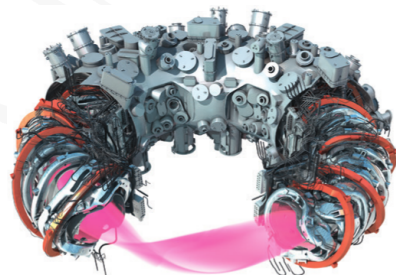
WENDELSTEIN 7-X STELLARATOR

- World's largest stellarator ($R=5.5$ m, $a=0.53$ m)
- 70 superconducting magnets: 50 non-planar + 20 planar coils
- Volume: 30 m^3 , $B_{\text{max}} = 3\text{ T}$
- Island divertor configuration (5/6, 5/5, 5/4)
- Optimized for reduced neoclassical transport
- Heating power: up to 14 MW ECRH
- Operational phases: OP1.1, OP1.2a, OP1.2b
- Achieved 30 min plasma discharges



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Key Challenge

Managing heat loads on plasma-facing components through controlled radiative cooling

PLASMA RADIATION AND TRANSPORT

Transport Mechanisms:

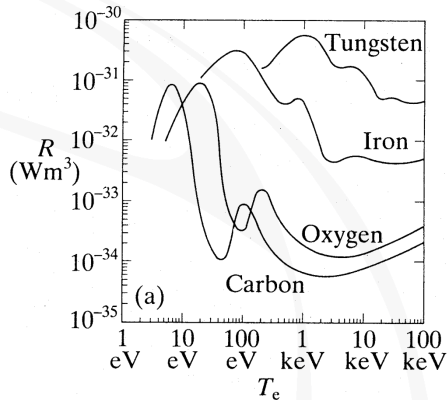
- Classical: collisional diffusion
- Neoclassical: trapped particles, $\propto 1/\nu$
- Anomalous: turbulent transport (dominant)

Radiation Processes:

- Bremsstrahlung: $P_{\text{Brems}} \propto n_e^2 T^{1/2}$
- Line radiation: atomic transitions
- Impurity effects on plasma performance

Radiation Fraction:

$$f = \frac{P_{\text{rad}}}{P_{\text{heat}}}$$



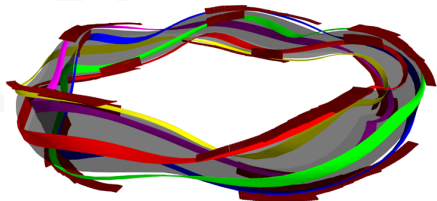
Impurity Seeding

Deliberate injection of low-Z (He, N₂) or high-Z (Ne, Ar) impurities to enhance

ISLAND DIVERTOR CONFIGURATION

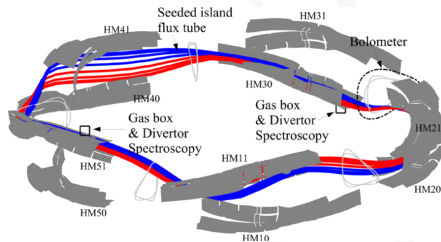
Magnetic Configuration:

- Standard configuration: $\iota = 5/5$
- Five helical island chains
- Ten divertor modules (5 half-modules)
- Field lines connect upstream to downstream targets
- Connection lengths: $\mathcal{O}(100 \text{ m to } 1000 \text{ m})$



Gas Injection Strategy:

- Thermal helium beam valves
- Located behind divertor targets
- Direct seeding into island chains
- Fast piezo valves (ms response)

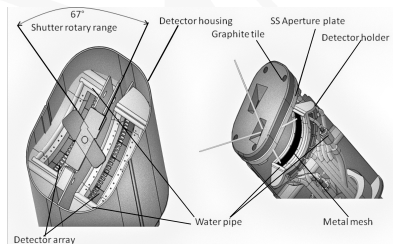


BOLOMETER DIAGNOSTIC SYSTEM

BOLOMETRY AT W7-X

Metal Resistor Bolometers:

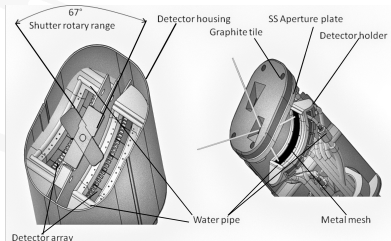
- Thin gold film absorbers ($5\text{ }\mu\text{m}$ thickness)
- Wheatstone bridge circuit configuration
- Measures total radiation power (broadband)
- Multicamera system: HBC (32 ch), VBCI/r (2×24 ch)
- 80 channels total (61 functional in OP1.2b)
- Spatial resolution: $\sim 5\text{ cm}$ at magnetic axis
- Unity absorption efficiency: 600-0.2 nm



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- 80 channels total (61 functional in OP1.2b)
- Spatial resolution: ~ 5 cm at magnetic axis
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Measurement Principle:

$$P_M = F_M \cdot \left(\frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{P,\text{tor}}}{V_C} \sum_M \frac{P_M V_M}{A_M}$$

BOLOMETER CALIBRATION AND PERFORMANCE

Calibration Methods:

- In-situ laser calibration
- Electrical calibration
- Etendue calculations:
$$K_M = \int_M \widetilde{K}_M dA_M$$
- Volume determination: V_M (LOS cone volume)

Performance Characteristics:

- Time resolution: 0.8 ms minimum (Δt 1.6 ms for feedback)
- Sensitivity: $\sim 10 \text{ kW m}^{-3}$
- Measurement uncertainty: $\sim 10\text{-}15\%$
- In-situ ohmic heating calibration

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Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information

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Applications:

- Power balance calculations
- Radiation profile monitoring
- Real-time feedback control
- Tomographic reconstruction

REAL-TIME RADIATION FEED- BACK CONTROL

FEEDBACK SYSTEM DESIGN

System Architecture:

- NI 6321 data acquisition hardware
- LabVIEW control software
- PID controller implementation
- Thermal gas valve actuation
- Minimum latency: 13.6 ms

PID Controller:

$$u(t) = K_p e(t) + K_i \int e(t') dt' + K_d \frac{de(t)}{dt}$$

Typical parameters: $K_p = 0.5-2.0$, $K_i = 0.1-0.5$, $K_d = 0$

Radiation Prediction Proxies:

Multi-channel (3-7 channels):

$$P_{\text{pred}}^{(1)} = \frac{V_{P,\text{tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

Validated subset: $S = \{65, 78, 61, 73, 86\}$ (VBC)

Single channel (fast):

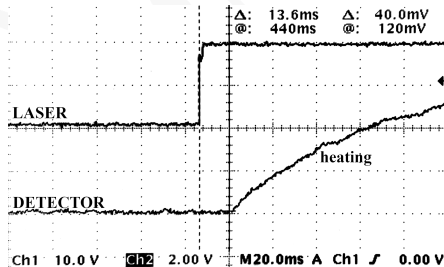
$$P_{\text{pred}}^{(2)} \propto \Delta U_M$$

Dimensionless, minimal processing overhead

FEEDBACK SYSTEM DESIGN (CONT.)

System Performance:

- Minimum acquisition latency: 13.6 ms
- FIFO smoothing: $\sim M \cdot \Delta t / 2$ ($M=10$, $\Delta t=1.6$ ms)
- Total system latency: 200 – 400 ms (incl. plasma response)
- Competitive with other diagnostics
- Benchmark testing with laser diode validated timing



Latency measurement with laser diode

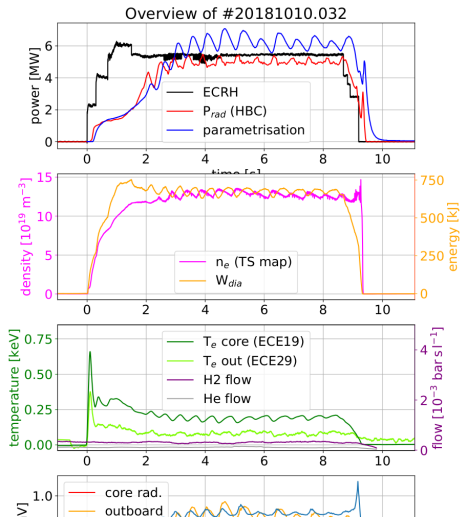
Operational Experience:

- Deployed during OP1.2b campaign (2018)
- Stable operation over multiple discharges
- Robust to plasma variations and transients

EXPERIMENTAL ACHIEVEMENTS

XP20181010.32 - Benchmark Discharge:

- Duration: 9.2 s, $P_{\text{ECRH}} = 6.23$ MW (O2-mode)
- Total input energy: 55 MJ
- Stable hydrogen-seeded detachment
- Radiation fraction: $f_{\text{rad}} > 90\%$ sustained
- Peak values up to 100%
- Target heat load reduction: factor > 2
- C^{3+} detachment visible at $f_{\text{rad}} \sim 50\%$
- Validated 5-channel VBC subset ($S = \{65, 78, 61, 73, 86\}$)



EXPERIMENTAL ACHIEVEMENTS (CONT.)

Comparison with Other Feedback Methods:

Electron Density Feedback:

- Indirect control of radiation
- Density affects confinement
- Less direct connection to detachment

C-III Filterscope Feedback:

- Spectroscopic proxy
- Single impurity species
- Limited spatial information

Bolometer Feedback:

- Direct radiation measurement
- Intrinsic connection to detachment physics
- Total power control
- Multiple viewing chords
- Robust to impurity mix changes

Advantage

Bolometer feedback directly controls the relevant quantity (radiation) unlike density or spectroscopy proxies

FEEDBACK PERFORMANCE ANALYSIS

- **Gas Injection Dynamics:** Moderately scaled gas puffs (medium length and intensity) most effective for reliable edge cooling
- **System Latency:** Total 200 – 400 ms (measurement + plasma response)
 - Minimum acquisition latency: 13.6 ms
 - FIFO smoothing adds $\sim M \cdot \Delta t / 2$ delay ($M=10$, $\Delta t=1.6$ ms)
- **Optimization Challenges:** Difficult to optimize during commissioning due to limited experimental time and complex parameter space

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Critical Finding

The intrinsic connection of P_{rad} and f_{rad} to the detachment process makes bolometer feedback essential for future fusion reactor applications

LINE OF SIGHT SENSITIVITY AND MODELING

IMPURITY SEEDING MODELS

Two-Chamber Model:

$$\dot{N}_w = (N_{w,\text{lim}} - N_w) \tau_{w,p}$$

$$\dot{N}_p = \Gamma_s + N_w \tau_{w,p} - f_{w,p} N_p - N_p \tau_p$$

- Plasma and wall compartments
- Particle exchange rates
- Pumping and recycling

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Three-Chamber Model:

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$$\dot{N}_s = \Gamma_s - N_s (\tau_{s,p} + \tau_s) + g(N_p, N_s) - c_0$$

$$\dot{N}_p = N_s \tau_{s,p} - \frac{N_p N_s}{N_{p,\text{lim}}}$$

- Adds SOL compartment
- Better represents edge physics
- Both models equally capable of representing feedback measurements

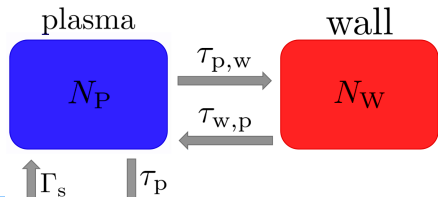
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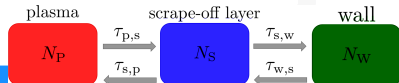
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CHANNEL SELECTION SENSITIVITY

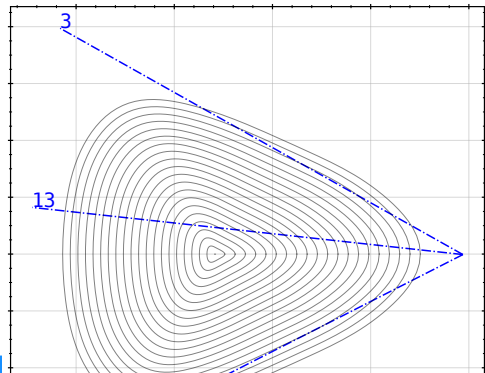
LOS Sensitivity Evaluation:

- Weighted deviation:

$$\Delta_w = \sum_i w_i |\Delta P_i| / \sum_i w_i$$
- Correlation coefficient: r (Pearson)
- Mean deviation: $\bar{\Delta} = \langle |\Delta P_i| \rangle$
- Evaluated across 45 discharge pairs from OP1.2b
- Threshold: $\Delta_w < 0.15$ for acceptable performance

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves $\geq 85\%$ prediction accuracy



STRAHL IMPURITY TRANSPORT MODELING

STRAHL Code:

- 1D impurity transport simulation
- Models carbon and oxygen radiation
- Includes ionization, recombination
- Diffusion and convection transport

Key Results:

- Carbon dominates SOL emissivity (C^{3+} , C^{4+})
- Reduced diffusivity near separatrix: $D_{\perp}$
 $0.1 \text{ m}^2 \text{ s}^{-1}$
- Inward shift of radiation for $f_{\text{rad}} \rightarrow 1$
(toward $\rho_{\text{pol}} \approx 0.95$)
- Radiation moves from outside to inside

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Forward Modeling:

- Calculate chord brightness from STRAHL profiles
- Compare with experimental measurements
- Validate transport assumptions

Discrepancy

Forward calculations show less asymmetry in chord brightness than experiments, suggesting model limitations in capturing 3D effects

TOMOGRAPHIC RECONSTRUCTION

MINIMUM FISHER REGULARIZATION (MFR)

Ill-Posed Inverse Problem:

- 128 line-integrated measurements
- Reconstruct 2D emissivity: ~ 4500 pixels
- Requires regularization

Fisher Information:

$$I_F = \int \frac{1}{g(\vec{r})} \left(\frac{\partial g(\vec{r})}{\partial \vec{r}} \right)^2 d\vec{r}$$

Iterative Solution:

Radially Dependent Anisotropy (RDA):

- Tailored weighting:
 $\alpha(\rho) = \alpha_0 + (\alpha_1 - \alpha_0) \cdot \rho^\beta$
- $\alpha < 1$: localized structures (edge)
- $\alpha > 1$: smooth distributions (core)
- Typical: $\alpha_0 \approx 0.1$, $\alpha_1 \approx 10$, $\beta \approx 2$
- Accounts for stellarator flux surface geometry

MFR Advantages

- Robust to noisy data
- Smooth solutions
- Incorporates a priori knowledge

GEOMETRY SENSITIVITY AND BENCHMARKING

Geometry Perturbation Tests:

- Camera position variations
- Detector/aperture segmentation
($N=2,4,8$)
- Triangulation vs rectangular splitting
- Etendue calculation validation

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Key Findings:

- Significant robustness to geometry variations
- $N \geq 8$ segmentation adequate
- Intrinsic bias toward upper SOL/separatrix
- Camera displacement effects non-negligible

GEOMETRY SENSITIVITY AND BENCHMARKING

Phantom Image Benchmarks:

- 12 synthetic test cases (Gaussian, ring, asymmetric)
- Quality metrics: χ^2 , ρ_c
- Noise sensitivity: up to 10% Gaussian noise
- Geometry perturbations: $\pm 1^\circ$ camera tilts
- Reconstruction quality $> 90\%$ for well-constrained cases
- Optimal k_{ani} depends on profile

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Recommended Settings

$k_{\text{ani}} = \{2.0, 0.3\}$ (core, edge)

Grid: 30×150 pixels

Domain: $1.3^2 V_P$

EXPERIMENTAL TOMOGRAPHY RESULTS

Statistical Analysis:

- Forward vs backward P_{rad} correlation
- Tomographic integral:
$$P_{\text{rad,tomo}} = \iiint \epsilon(\mathbf{r}) dV$$
- Agreement with bolometer sum within 10-20%
- Core radiation slightly overestimated
- Reveals SOL radiation underestimation in some cases
- Consistent with phantom benchmarks

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Power Balance Application:

$$P_{\text{bal}} = P_{\text{ECRH}} - P_{\text{rad}} - P_{\text{div}}$$

- $P_{2\text{D}}^{\text{core}}$ improves balance accuracy
- More stable than P_{rad} alone
- Alternative for quasi-steady-state analysis

TOMOGRAPHY QUALITY ASSESSMENT

- **Reconstruction Accuracy:** P_{rad} consistently underestimates total 2D integrated power by amount equal to SOL radiation
- **Quality Metrics:** χ^2 and correlation coefficient ρ_c not always congruent - optimal tomogram may not have $\chi^2 = 1$
- **Anisotropy Optimization:** Ideal k_{ani} set exists for each profile but dimensionality makes finding it impractical
- **Experimental Validation:** Results agree well with phantom benchmarks, supporting algorithm reliability

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Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for physics analysis and power balance calculations, though computational limitations prevent continuous real-time operation

CONCLUSIONS AND OUTLOOK

KEY ACHIEVEMENTS

1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
2. **Line of Sight Optimization:** Systematic sensitivity analysis identified optimal detector configurations
3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting

KEY ACHIEVEMENTS

1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
 - Achieved stable detachment with $f_{\text{rad}} \geq 85\%$, peaks up to 100%
 - Reduced target heat loads by factor >2
 - Minimum system latency: 13.6 ms + 200 – 400 ms plasma response
 - Validated 3-5 channel LOS subsets for fast prediction (85% accuracy)
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3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting
 - Robust to geometry perturbations ($\pm 1^\circ$) and noise (10%)
 - Successfully applied to experimental data (OP1.2b)
 - Reconstruction quality $>90\%$ for well-constrained phantoms

SCIENTIFIC INSIGHTS

Detachment Physics:

- C^{3+} detachment visible at $f_{\text{rad}} \sim 50\%$
- Stable detachment achievable at $f_{\text{rad}} > 90\%$
- Radiation shifts inward as $f_{\text{rad}} \rightarrow 1$
- X-point and island radiation concentration

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- Intrinsic connection to detachment essential
- System latency limits performance

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Diagnostic Performance:

- Multicamera system provides comprehensive coverage
- Geometry uncertainties affect results
- Tomography reveals SOL radiation underestimation

FUTURE OUTLOOK AND RECOMMENDATIONS

System Upgrades:

- Reduce feedback latency: target <10 ms
- Implement more complex P_{pred} models (machine learning)
- Per-experiment configurability and auto-tuning
- Develop predictive scaling laws for gas injection
- Integration with plasma control system (PCS)

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Tomography Extensions:

- Larger phantom benchmark set (>50 cases)
- Additional virtual camera arrays (test future upgrades)
- Real-time capable algorithms (GPU acceleration)
- Automated parameter optimization (grid search, ML)

FUTURE OUTLOOK AND RECOMMENDATIONS

System Upgrades:

- Reduce feedback latency: target <10 ms
- Implement more complex P_{pred} models (machine learning)
- Per-experiment configurability and auto-tuning
- Develop predictive scaling laws for gas injection
- Integration with plasma control system (PCS)

Modeling Improvements:

- Refine multi-chamber models with

Tomography Extensions:

- Larger phantom benchmark set (>50 cases)
- Additional virtual camera arrays (test future upgrades)
- Real-time capable algorithms (GPU acceleration)
- Automated parameter optimization (grid search, ML)
- Integration with synthetic diagnostics

Reactor Relevance:

- Essential for DEMO power plant operation ($P_{\text{net}} > 500$ MW)

BROADER IMPACT

Contribution to Fusion Energy Research

This thesis advances the state-of-the-art in plasma diagnostics and control for stellarator fusion devices, providing essential tools and knowledge for achieving reactor-relevant operating scenarios with controlled radiative cooling and stable detachment.

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- First real-time bolometer feedback at W7-X
- Validated diagnostic optimization methods
- Benchmarked tomography algorithm
- Demonstrated stable high- f_{rad} operation

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- Validated diagnostic optimization methods
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- Demonstrated stable high- f_{rad} operation

Scientific Contributions:

- Detachment physics understanding
- Impurity transport insights
- Radiation distribution characterization
- Power balance methodology

SUMMARY

- Developed and operated first real-time bolometer radiation feedback system at W7-X stellarator
- Achieved stable, controlled detachment with radiation fractions $f_{\text{rad}} \geq 85\%$ and peaks up to 100%
- Reduced divertor target heat loads by factor of 2 through controlled impurity seeding
- Validated 3-channel bolometer subset for fast radiation power prediction with $\geq 85\%$ accuracy
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- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback
- STRAHL modeling revealed carbon dominance in SOL radiation and inward shift at high f_{rad}
- Developed and benchmarked Minimum Fisher Regularization tomography with radially dependent anisotropy
- Demonstrated robust 2D radiation reconstruction from experimental data
- Provided alternative power balance methodology using tomographic integrals

COLLABORATIONS

Acknowledgement

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Collaborators:



HELMHOLTZ
RESEARCH FOR GRAND CHALLENGES



EUROfusion

Thank You

REFERENCES I

Backup Slides

Additional Details

MEASUREMENT ALGORITHM OVERVIEW

Algorithm Stages:

1. **Initialization** (T1-60 s)
 - Load geometry matrices K_M, V_M
 - Configure channel selection S
 - Initialize analog output
2. **Calibration** (4 s)
 - Offset measurement
 - Two-stage calibration
 - Determine R, τ, κ
3. **Measurement**
 - Continuous voltage integration
 - Moving average filtering
 - Real-time prediction

Real-time Feedback:

- Parallel analog output via NI 6321
- Two prediction channels $P_{\text{pred}}^{(1)}, P_{\text{pred}}^{(2)}$
- FIFO buffer management
- Sample time Δt configurable

Data Storage:

- Local HDD backup
- W7-X archive upload
- Calibration parameters
- Full voltage traces

FOURIER TRANSFORM CORRELATION METRIC

Alternative LOS Sensitivity Evaluation:

Method:

- Uses spectral correlation between P_{rad} and P_{pred}
- Discrete Fast Fourier Transform (DFFT)
- Cross-spectral density analysis
- Coherence-based metric

Mathematical Formulation:

$$\gamma_i = \mathcal{F}(P_{\text{rad}})(\omega)$$

$$\gamma_j = \mathcal{F}(P_{\text{pred}})(\omega)$$

$$\varphi(\omega) = \sqrt{\frac{|g_{i,j}^2|}{|\gamma_i \gamma_j|}}$$

Key Results:

- Similar patterns to correlation metric
- Global maximum at upper inboard X-point
- HBC: best at $0.38r_a$
- VBC: best at $-0.13r_a$
- Spectral focus excludes absolute value contributions

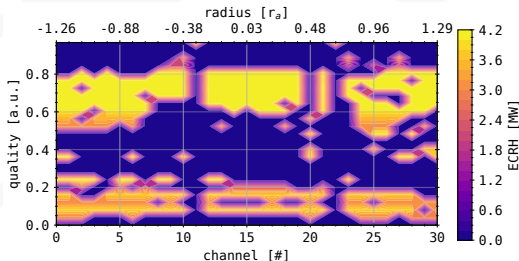
Advantages:

- Frequency domain analysis

Robust to noise

PLASMA PARAMETER SENSITIVITY ANALYSIS

Correlation with Plasma Parameters:



Heating power P_{ECRH} correlation

Key Findings:

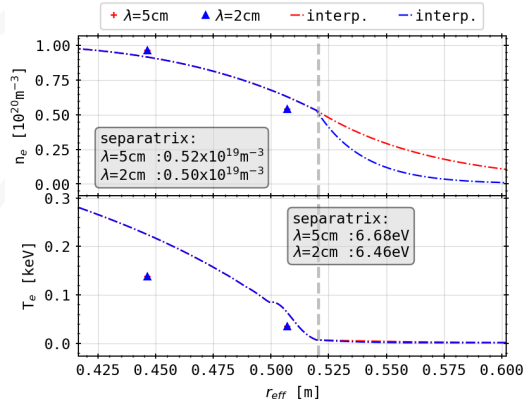
- Higher P_{ECRH} (≥ 3.5 MW) $\rightarrow$ better predictability
- Optimal $f_{\text{rad}} > 0.75$
- Core $T_e \sim 1.3$ keV ideal
- Gap at $\vartheta \approx 0.3-0.5$
- Both cameras in good agreement

Optimal Discharge Conditions:

- Medium-high heating power
- High radiation fraction
- Moderate core temperature
- Consistent across LOS combinations

STRAHL PARAMETER VARIATIONS

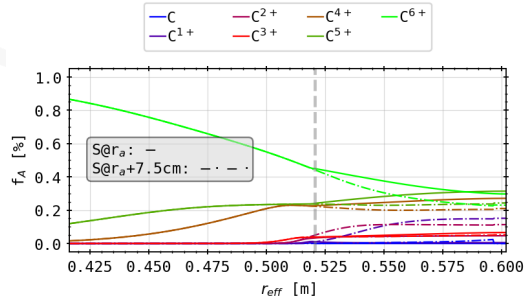
Decay Length Variation:



SOL decay length: $\lambda = 2\text{ cm}$ vs 5 cm

Impact:

Impurity Source Location:

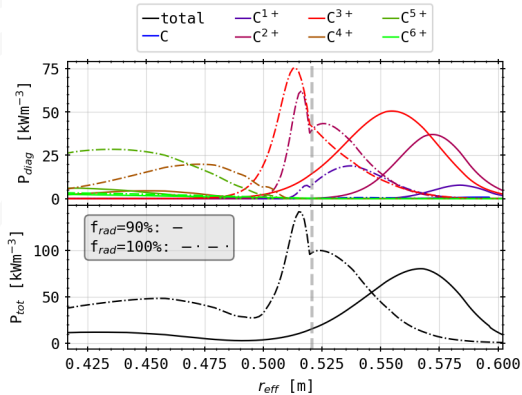


Source at LCFS vs divertor ($r_a+7.5\text{ cm}$)

Impact:

- Changes ion populations in SOL
- Similar core emissivity results

STRAHL: RADIATION FRACTION DEPENDENCE



Total radiation for $f_{\text{rad}} = 90\%$ and 100%

Critical Observations:

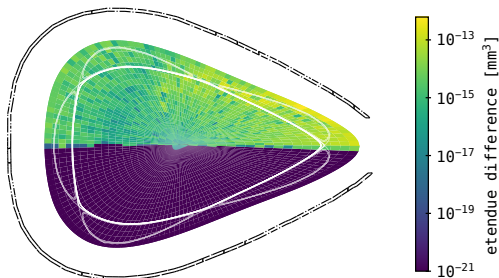
- At $f_{\text{rad}} = 90\%$: peak in SOL
- At $f_{\text{rad}} = 100\%$: peak shifts to LCFS
- C^{3+} and C^{2+} most sensitive
- Dramatic profile changes near separatrix

Implications:

- LOS sensitivity strongly depends on f_{rad}
- Detachment scenarios predictable
- Validates feedback system design
- Critical for controlled detachment

CAMERA GEOMETRY SENSITIVITY

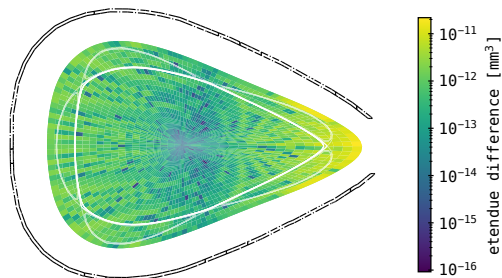
Unilateral Mirroring:



Etendue change: $<10^{-13}$ permm

- Half of HBC array mirrored
- Minimal impact on forward model
- Asymmetry reduced by 0.5-1%

Artificial Symmetric Camera:



Fully symmetric, equidistant array

- Perfectly symmetric geometry
- Similar etendue variations
- Validates as built design

TOMOGRAPHY: RDA VS RGS COMPARISON

Radially Dependent Anisotropy (RDA):

- Standard MFR with k_{ani} profile
- Smooth reconstructions
- Good radial profile matching
- Established method

Relative Gradient Smoothing (RGS):

- Weight: $W_{\text{RGS}} = (1/g_i)^2$
- Can be combined with RDA
- Stronger feature localization
- Higher local intensities

Comparison Results:

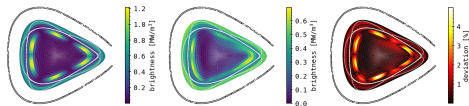
- Similar integrated power values
- RGS: sharper features
- RGS: better peak localization
- RGS: may lose weak structures
- RDA: more robust for general use

Conclusion:

- RDA sufficient for most cases
- RGS useful for specific scenarios
- Combined approach possible
- Choice depends on physics goals

PHANTOM TOMOGRAPHY: EDGE CASES

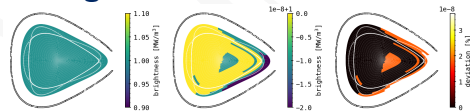
Inverted Anisotropy:



Bright SOL ring + core islands

- More challenging reconstruction
- SOL dominates measurement
- Core features partially obscured

Homogeneous Distribution:



Uniform 1 MW m^{-3} everywhere

- Tests baseline reconstruction
- Validates algorithm behavior
- Confirms no artificial structures

Lessons: Camera sensitivity in SOL can obscure core features; algorithm performs well for uniform distributions; inverted scenarios more challenging than standard core-peaked profiles.

APPENDIX SUMMARY

Measurement Algorithm:

- Three-stage process validated
- Real-time capability confirmed
- Robust calibration procedure
- Full data archiving

LOS Sensitivity Methods:

- Fourier correlation complements time-domain
- Plasma parameter correlations identified
- Optimal discharge conditions determined

STRAHL Validation:

- Parameter variations tested
- Model robustness confirmed
- Critical f_{rad} dependence shown

Tomography Extensions:

- Geometry perturbations minimal
- RDA vs RGS compared
- Edge case phantoms tested
- Algorithm limits characterized